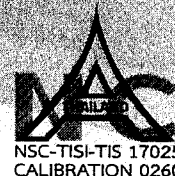


CRYSTAL CALIBRATION SALES AND SERVICE CO., LTD.

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NSC-TISI-TIS 17025
CALIBRATION 0260

CERTIFICATE OF CALIBRATION

Certificate No. : 25-1390-079

Issue Date : 5 November 2025

Work Order No. : 25/1390

Customer Name : กลุ่มงานเทคนิคการแพทย์ โรงพยาบาลพระจอมเกล้า จังหวัดเพชรบุรี
53 ถ.รถไฟ ต.คลองกระแซง
อ.เมือง จ.เพชรบุรี 76000

Date of Received : 27 October 2025

Date of Calibration : 27 October 2025

Instrument Details : Description : Temperature Controlled Enclosures [Refrigerator]
Manufacturer : Sunhui
Model : LD2 DC 70
Serial No. : 5341
ID No. : 4110-007-0004-1/55
Resolution : 1 °C
Location : Special Lab

Calibration Method : This instrument was calibrated by insert standard thermometer into the chamber according to calibration procedure no. CWI-T-10 follow up to TLAS G-20-1/02-08
(E) : Guidelines for Calibration and Checks of Temperature Controlled Enclosures.

Environmental Conditions :

Temperature : Area Monitoring between 15°C to 40°C

Humidity : Area Monitoring between 30%RH to 85%RH

Line Voltage : Area Monitoring 220 VAC $\pm$ 10%

Traceability of Measurement :

This certificate of calibration documents the traceability to national standard, which realize the unit of measurement according to the International system of Units (SI) and The temperature scale in use at this laboratory is The International Temperature scale of 1990.

Calibrated by : Mr. Kritsada Kaewwangpa
Calibration Engineer

Approved by : 
(Mr. Anuwat Yaklermjit)
Laboratory Manager

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CERTIFICATE OF CALIBRATION

Certificate No. : 25-1390-079

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Details of Calibration

1. Reference Standards Instrument

Instrument	Model	Serial No./Ins No.	Certificate No.	Due Date
Data Acquisition unit	34972A	MY57001206	25-0861-021	04 July 2026
Sensor type	RTD	CH # 201-209	25-0861-021	04 July 2026

2. Certificate traceble : This certificate traceable to The International System of Unit refer to
Crystal Calibration Sales and Service Co., Ltd. , NSC-ONSC Calibration No. 0260

3. Condition of item : Used

4. Calibration site : On - Site

5. Result of Calibration : Without adjustment

6. Evaluate Condition : Time Constant : - Hour 33 Minute At cal. point 4 °C
Air vent : Off
Fan speed status : Fixed Fan Speed

7. Calibration note : The results reported in this certificate refer to the condition of instrument on
the process into the steady state of chamber

8. Sensors Installation Diagram : When ; Sensor installation location in Chamber @ Working Space
A = Distance between sensor and wall of chamber is 10 cm

9. Dimensions of chamber : W = 1 m ; D = 0.5 m ; H = 1.2 m

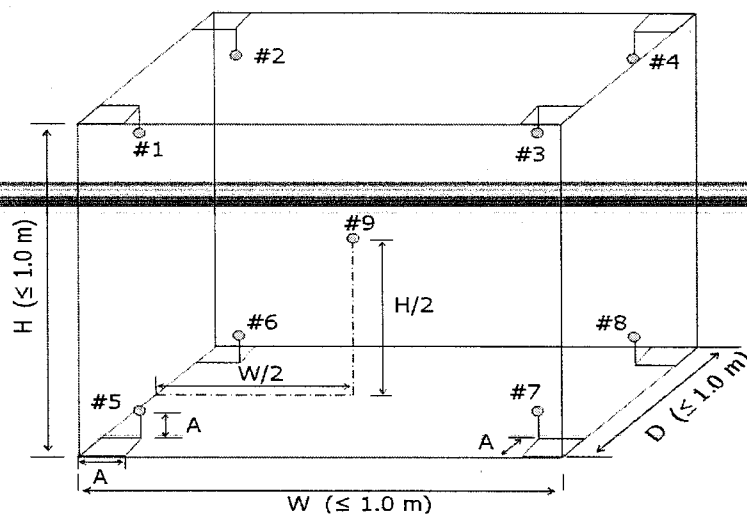
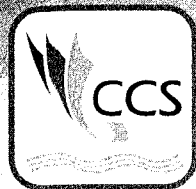


Diagram of Chamber



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CERTIFICATE OF CALIBRATION

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Result of Temperature Distribution and Performance Check

Table1 : Reporting of Temperature Distribution

Calibration point (°C)	Average Measured Temperature (°C) @ Sensor No. (Sensor No.9 is REF)									Uncertainty ± (°C)
	#1	#2	#3	#4	#5	#6	#7	#8	#9	
4	5.10	5.44	4.74	4.17	5.83	6.43	3.48	3.44	4.44	1.9

Table 2 : Reporting of Performance check

Calibration Point (°C)	Indicator Set Point (°C)	Indicator Reading (°C)			Stability ± (°C)	Uniformity (°C)	Overall variation (°C)
		MAX	MIN	Average			
4	4	5	4	4	1.34	2.72	5.99

Note

Customer can't removed items form chamber and this report customer request and accepted in certificate

The reference sensor is preferably located of the geometric center of chamber

The measured temperature data readout by software "Benchlink Datalogger 3"

The quoted uncertainty include " Stability " and " Loading effect (20% of Temp Uniformity) "

Stability - one-half of the greatest maximum difference of measured temperatures at any one sensor.

Uniformity - the maximum difference of measured temperatures at any sensors and the measured temperature at the reference location which are observed at the same time or at as close an observation time as possible to determine the temperature pattern or homogeneity within the chamber under steady state conditions.

Overall Variation - The difference of the maximum and minimum measured temperatures througout observation time.

Indicating Temperature - the average reading of indicating device that forms the integral part of the enclosure.

This result of calibration was found accurate as shown on date and place of calibration only.

The reported uncertainty of measurement was based on a standard uncertainty multiplied by a coverage factor $k=2$ providing a level of confidence of approximately 95%.

-END-

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Statements of conformity report

Refer to Certificate No. : 25-1390-079

Issue Date : 5 November 2025

Work Order No. : 25/1390

Detail of Equipment

Description : Temperature Controlled Enclosures [Refrigerator]

Manufacturer : Sunhui

ID No. : 4110-007-0004-1/55

Model : LD2 DC 70

Serial No. : 5341

Result of Calibration

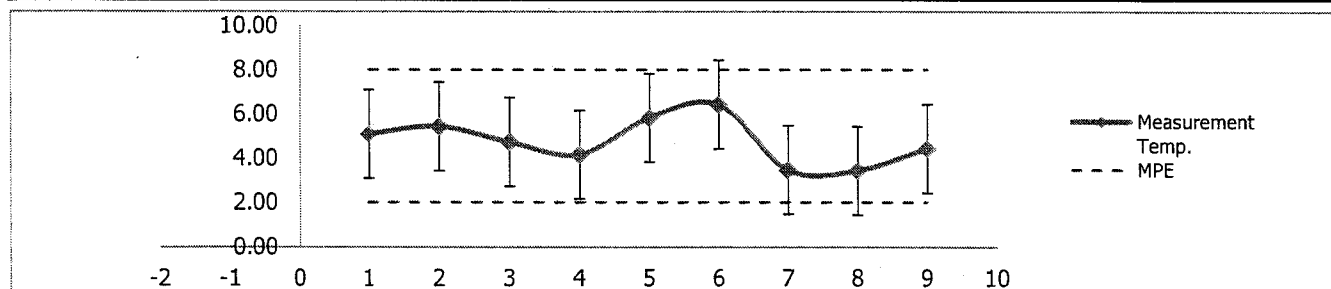
This result of calibration : Without adjustment

Resolution : 1 °C

Calibration point	4	°C
Customer MPE	2-8	°C

Uncertainty		Uncertainty MPE		Evaluate resulte
1.9	°C	3.0	°C	Passed

Evaluate upper limit [UL]				Evaluate lower limit [LL]			
Position	MEAN STD	MPE	Lab. Decide	Position	MEAN STD	MPE	Lab. Decide
#1	5.10	8.0	Pass	#1	5.10	2.0	Pass
#2	5.44	8.0	Pass	#2	5.44	2.0	Pass
#3	4.74	8.0	Pass	#3	4.74	2.0	Pass
#4	4.17	8.0	Pass	#4	4.17	2.0	Pass
#5	5.83	8.0	Pass	#5	5.83	2.0	Pass
#6	6.43	8.0	Pass	#6	6.43	2.0	Pass
#7	3.48	8.0	Pass	#7	3.48	2.0	Pass
#8	3.44	8.0	Pass	#8	3.44	2.0	Pass
#9	4.44	8.0	Pass	#9	4.44	2.0	Pass



This statement judgment and decision by Laboratory guideline only

The conformity certificate documents validity following ISO/IEC Guide 98-4 : Role of measurement uncertainty in conformity assessment based on statement with guard band refer to AS 3864 consider expanded measurement uncertainty (k=95%)

The tolerance and decision rules ;

Decision rules : Laboratory decide temperature measurement in chamber is ;

MPE of Customer or Lab decide = Measurement temperature of any points ; [$\leq$ MPE is pass , $>$ MPE is Fail]

Statements of conformity decision by :

(Mr. Anuwat Yaklermjit)

Laboratory Manager

This document is a guideline for interpreting calibration report for customers only is not part documentation ISO/IEC 17025:2017